

Backend API Specification

Branch Manager Module · Proposed REST API Contract

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1. Overview & Scope

1.1 Document Purpose

This document is the authoritative backend API specification for the **Fixed Assets** feature within the Branch Manager mobile application. The feature currently uses mock data inside Flutter repositories and no real backend endpoints are wired. This specification proposes, defines, and formalises all required REST endpoints, request/response contracts, and business rules inferred from the Flutter implementation and screen flows.

■ All endpoints, request shapes, and response structures described in this document are **proposed**. The PHP backend team should treat this as the implementation target and raise any conflicts with the mobile team before development begins.

1.2 Screens Covered

Screen	Class Name	Backend Required
Fixed Assets Dashboard	<code>FixedAssetsScreen</code>	Yes
Search	<code>FixedAssetsSearchScreen</code>	Yes
Search Results	<code>FixedAssetsSearchResultScreen</code>	Yes
Receive Assets	<code>FixedAssetsReceiveAssetsScreen</code>	Yes
Transfers & Disposal	<code>FixedAssetsTransfersAndDisposalScreen</code>	Yes
Requests (Modifications / Transfers / Disposals)	<code>FixedAssetsRequestsScreen</code>	Yes
Modification Details	<code>FixedAssetsModificationDetailsScreen</code>	Yes
Modify Asset Status	<code>FixedAssetsModifyAssetStatusScreen</code>	Yes
Success Screens	—	No (no backend call)

2. API Conventions

2.1 Base URL & Authentication

All Fixed Assets endpoints share a common base path:

```
Base Path: /branch-manager/fixed-assets
```

Authentication uses the same **Bearer Token** mechanism as all other Branch Manager endpoints. The token must be passed in the **Authorization** HTTP header on every request.

```
Authorization: Bearer {access_token}
```

2.2 Content Types & Encoding

Request Type	Content-Type	Used For
Standard requests	application/json	GET, POST with JSON body
File uploads	multipart/form-data	POST / PATCH with images or files

2.3 Timestamps & Enum Formats

All timestamps must be in **ISO-8601** format (e.g. 2024-06-01T12:00:00Z). Enum/status values use **lowercase snake_case** to match the Dart enum definitions in the Flutter client (e.g. need_attention, transfer_to_branch).

3. Shared Response Objects

The following objects are reused across multiple endpoints. All backend responses must match these shapes exactly.

3.1 AssetLocationResponse

Represents a physical location (zone/branch) within the system.

```
{
  "id": "string",
  "name": "string"
}
```

Field	Type	Required	Description / Validation Rules
id	string	Yes	Unique identifier for the location.
name	string	Yes	Human-readable location name.

3.2 AssetSearchItemResponse

The primary asset record returned in search results, asset lists, and picker endpoints. Used extensively across the feature.

```
{
  "id": "string",
  "name": "string",
  "code": "string",
  "image": "string",
  "location": { "id": "string", "name": "string" },
  "status": "excellent | need_attention | problem",
  "assigned_to": "string",
  "age": "string",
  "value": "string",
  "custody": "string"
}
```

Field	Type	Required	Description / Validation Rules
id	string	Yes	Unique asset identifier.
name	string	Yes	Human-readable asset name.
code	string	Yes	Asset code / barcode identifier.
image	string	Yes	URL of asset image. May be empty string if no image is available.
location	AssetLocationResponse	Yes	Nested location object.

Field	Type	Required	Description / Validation Rules
status	enum	Yes	One of: excellent, need_attention, problem.
assigned_to	string	Yes	Name of the employee the asset is assigned to.
age	string	Yes	Readable age text, e.g. '6 Months'. UI displays as-is.
value	string	Yes	Asset monetary value as numeric text (e.g. '12500').
custody	string	Yes	Custody duration text, e.g. '2 Months'. UI displays as-is.

3.3 ReceiveAssetsItemResponse

Represents a pending asset awaiting receipt by the branch.

```
{
  "id": "string",
  "assetName": "string",
  "assetCode": "string",
  "assetImage": "string"
}
```

■ Note: This object uses camelCase keys (assetName, assetCode, assetImage) rather than snake_case. This inconsistency exists in the Flutter contract. The backend must match this casing exactly for this endpoint.

Field	Type	Required	Description / Validation Rules
id	string	Yes	Unique asset identifier for the pending receipt.
assetName	string	Yes	Asset name (camelCase key required).
assetCode	string	Yes	Asset code (camelCase key required).
assetImage	string	Yes	Asset image URL. May be empty string.

3.4 Supporting Shared Objects

Lightweight reference objects used in lookups and pickers:

```
// ZoneResponse
{ "id": "string", "name": "string" }

// TypeResponse
{ "id": "string", "name": "string" }

// ResponsibleEmployeeResponse
{ "id": "string", "name": "string", "image": "string" }

// BranchResponse
{ "id": "string", "name": "string", "image": "string" }
```

3.5 PaginationMetaResponse

Standard pagination metadata returned alongside paginated lists.

```
{
  "current_page": 1,
  "last_page": 10,
  "total": 97
}
```

Field	Type	Required	Description / Validation Rules
current_page	int	Yes	The current page number (1-indexed).
last_page	int	Yes	The index of the last available page.
total	int	Yes	Total number of records across all pages.

3.6 AttachmentResponse

Represents an uploaded file attachment (photos, documents).

```
{
  "id": "string",
  "file_name": "string",
  "file_type": "string",
  "file_size": 0,
  "url": "string",
  "uploaded_at": "2024-06-01T12:00:00Z"
}
```

Field	Type	Required	Description / Validation Rules
id	string	Yes	Unique attachment identifier.
file_name	string	Yes	Original file name.
file_type	string	Yes	MIME type, e.g. image/jpeg.
file_size	int	Yes	File size in bytes.
url	string	Yes	Publicly accessible download URL.
uploaded_at	ISO-8601	Yes	Upload timestamp.

3.7 TimelineItemResponse & TimelineResponse

Timeline items track the lifecycle events of a request (submitted, reviewed, approved, etc.).

```
{
  "timelines": [
    {
      "id": "string",
      "event_type": "string",
      "name": "string",
      "image": "string",
      "occurred_at": "2024-06-01T12:00:00Z"
    }
  ]
}
```

Field	Type	Required	Description / Validation Rules
id	string	Yes	Unique timeline event identifier.
event_type	string	Yes	Must match TimeLineEnum values on the backend.
name	string	Yes	Description of the event, e.g. 'Request Submitted'.
image	string	optional	URL of actor avatar; may be null or empty string.
occurred_at	ISO-8601	Yes	When this event occurred.

4. Detailed Response Contracts

4.1 SearchSummaryResponse & SearchResponse

Returned from the Search Assets endpoint. The root shape nests a **summary** block alongside the paginated **items** list inside a data wrapper.

```
{
  "asset_type_name": "Coffee Machines",
  "data": {
    "summary": {
      "total_items": 20,
      "total_items_matching_search": 5,
      "total_items_matching_search_excellent_status": 2,
      "total_items_matching_search_maintenance_status": 2,
      "total_items_matching_search_problem_status": 1
    },
    "items": [
      {
        "id": "string",
        "name": "string",
        "code": "string",
        "image": "string",
        "location": { "id": "string", "name": "string" },
        "status": "excellent | need_attention | problem",
        "assigned_to": "string",
        "age": "string",
        "value": "string",
        "custody": "string"
      }
    ]
  }
}
```

■ The UI parses `data.summary` and `data.items` as distinct blocks. The backend must follow this exact nesting structure — any deviation will break the search results screen.

4.2 ModificationRequestsListResponse

```
{
  "data": [
    {
      "id": "string",
      "asset_name": "string",
      "status": "pending | approved"
    }
  ],
  "meta": {
    "current_page": 1,
    "last_page": 5,
    "total": 42
  }
}
```

4.3 TransfersRequestsListResponse

```
{
  "data": [
    {
      "id": "string",
      "asset_name": "string",
      "status": "pending | approved",
      "condition": "excellent | need_attention | problem",
      "date_and_time": "2024-06-01T12:00:00Z",
      "type": "from_branch | from_finance | to_branch"
    }
  ],
  "meta": {
    "current_page": 1,
    "last_page": 3,
    "total": 28
  }
}
```

4.4 DisposalsRequestsListResponse

```
{
  "data": [
    {
      "id": "string",
      "asset_name": "string",
      "status": "pending | approved",
      "date_and_time": "2024-06-01T12:00:00Z"
    }
  ],
  "meta": {
    "current_page": 1,
    "last_page": 2,
    "total": 15
  }
}
```

4.5 ModificationDetailsResponse

Full detail payload for a single modification request, including asset snapshot, modification metadata, required actions, and lifecycle timeline.

```

{
  "status": "pending | approved",
  "asset_details": {
    "id": "string",
    "name": "string",
    "code": "string",
    "image": "string",
    "location": { "id": "string", "name": "string" },
    "status": "excellent | need_attention | problem",
    "assigned_to": "string",
    "age": "string",
    "value": "string",
    "custody": "string"
  },
  "modification_details": {
    "new_status": "excellent | need_attention | problem",
    "reason": "string",
    "attachment": {
      "id": "string",
      "file_name": "string",
      "file_type": "string",
      "file_size": 1048576,
      "url": "string",
      "uploaded_at": "2024-06-01T12:00:00Z"
    }
  },
  "required_actions": {
    "done_actions": ["initial_check", "cleaning"],
    "next_action": "need_technician | halt | replace",
    "approval_request_owner_note": "string"
  },
  "timelines": []
}

```

Field	Type	Required	Description / Validation Rules
<code>status</code>	enum	Yes	Overall request status: pending or approved.
<code>asset_details</code>	AssetSearchItemResponse	Yes	Snapshot of the asset at time of request.
<code>modification_details.new_status</code>	enum	Yes	The proposed new status for the asset.
<code>modification_details.reason</code>	string	Yes	Textual reason for the modification.
<code>modification_details.attachment</code>	AttachmentResponse	Yes	Documentation photo attached to the request.
<code>required_actions.done_actions</code>	string[]	Yes	Array of completed actions: initial_check, cleaning.
<code>required_actions.next_action</code>	enum	Yes	Recommended next step: need_technician, halt, or replace.

Field	Type	Required	Description / Validation Rules
<code>required_actions.approval_request_owner_note</code>	string	Yes	Note from requestor to approver.
<code>timelines</code>	TimelineItem Response[]	Yes	Lifecycle event log. May be empty array.

4.6 FixedAssetsOverviewResponse

Dashboard summary counts for total assets split by status.

```
{
  "total": 84,
  "excellent": 78,
  "maintenance": 5,
  "problem": 1
}
```

Field	Type	Required	Description / Validation Rules
<code>total</code>	int	Yes	Total number of fixed assets in this branch.
<code>excellent</code>	int	Yes	Count of assets with excellent status.
<code>maintenance</code>	int	Yes	Count of assets under maintenance / need_attention.
<code>problem</code>	int	Yes	Count of assets with problem status.

4.7 FixedAssetsTransferDetailsResponse

Full detail payload for a single transfer request. Includes request metadata, the list of transferred assets with zone/type info, and a lifecycle timeline. The `skip_confirmation` flag controls whether the UI presents a confirmation step.

```

{
  "data": {
    "request_details": {
      "from": "string",
      "to": "string",
      "request_date": "string",
      "arrival_time": "string",
      "transfer_id": "string",
      "status": "string"
    },
    "assets": [
      {
        "name": "string",
        "code": "string",
        "image_url": "string",
        "zone": { "id": "string", "name": "string" },
        "type": { "id": "string", "name": "string" },
        "status": "excellent | need_attention | problem",
        "value": "string",
        "custody": "string",
        "transfer_reason": "string"
      }
    ],
    "skip_confirmation": false,
    "timelines": [
      {
        "id": "string",
        "event_type": "string",
        "name": "string",
        "image": "string",
        "occurred_at": "2024-06-01T12:00:00Z"
      }
    ]
  }
}

```

Field	Type	Required	Description / Validation Rules
<code>data.request_details.from</code>	string	Yes	Display name of the source branch/location.
<code>data.request_details.to</code>	string	Yes	Display name of the destination branch/location.
<code>data.request_details.request_date</code>	string	Yes	Formatted date string of when request was made.
<code>data.request_details.arrival_time</code>	string	Yes	Formatted expected arrival time string.
<code>data.request_details.transfer_id</code>	string	Yes	Human-readable transfer reference ID.
<code>data.request_details.status</code>	string	Yes	Maps to FixedAssetsRequestsStatusStrategy enum.
<code>data.assets[].zone</code>	ZoneResponse	Yes	Zone where asset is located. Uses shared ZoneResponse shape.

Field	Type	Required	Description / Validation Rules
<code>data.assets[].type</code>	TypeResponse	Yes	Asset type category. Uses shared TypeResponse shape.
<code>data.assets[].status</code>	enum	Yes	Asset condition: excellent, need_attention, or problem.
<code>data.assets[].transfer_reason</code>	string	Yes	Reason this asset is being transferred.
<code>data.skip_confirmation</code>	bool	Yes	If true, the UI skips the confirmation step entirely.
<code>data.timelines</code>	TimelineItemResponse[]	Yes	Lifecycle event log. May be empty array.

4.8 FixedAssetsDisposalsDetailsResponse

Full detail payload for a single disposal request. Includes disposal scheduling metadata, the list of assets marked for disposal, and a lifecycle timeline.

```
{
  "data": {
    "request_details": {
      "proposed_date": "string",
      "proposed_time": "string",
      "disposal_method": "string",
      "selected_assets": 0,
      "status": "string",
      "approved_on": "string | null"
    },
    "assets": [
      {
        "name": "string",
        "code": "string",
        "image_url": "string",
        "zone": { "id": "string", "name": "string" },
        "type": { "id": "string", "name": "string" },
        "status": "excellent | need_attention | problem",
        "value": "string",
        "custody": "string",
        "transfer_reason": "string"
      }
    ],
    "timelines": [
      {
        "id": "string",
        "event_type": "string",
        "name": "string",
        "image": "string",
        "occurred_at": "2024-06-01T12:00:00Z"
      }
    ]
  }
}
```

Field	Type	Required	Description / Validation Rules
<code>data.request_details.proposed_date</code>	string	Yes	Formatted date string for scheduled disposal.
<code>data.request_details.proposed_time</code>	string	Yes	Formatted time string for scheduled disposal.
<code>data.request_details.disposal_method</code>	string	Yes	One of: scrap, sell, return_to_supplier, other.
<code>data.request_details.selected_assets</code>	int	Yes	Total count of assets in this disposal request.
<code>data.request_details.status</code>	string	Yes	Maps to FixedAssetsRequestsStatusStrategy enum.
<code>data.request_details.approved_on</code>	string null	optional	Populated once approved. Null if still pending.
<code>data.assets[].zone</code>	ZoneResponse	Yes	Asset zone. Uses shared ZoneResponse shape.
<code>data.assets[].type</code>	TypeResponse	Yes	Asset type. Uses shared TypeResponse shape.
<code>data.assets[].status</code>	enum	Yes	Asset condition: excellent, need_attention, or problem.
<code>data.assets[].transfer_reason</code>	string	Yes	Disposal reason for this asset (maps to transfer_reason in Dart model).
<code>data.timelines</code>	TimelineItemResponse[]	Yes	Lifecycle event log. May be empty array.

5. Screen-by-Screen Flows

This section documents the backend data requirements for each screen in the Fixed Assets feature, as inferred from the Flutter UI implementation.

5.1 Fixed Assets Dashboard

Flutter class: `FixedAssetsScreen`

- Displays a summary count card showing total, excellent, maintenance, and problem assets.
- Quick action buttons navigate to: Search, Receive Assets, and Transfers/Disposal.
- Ongoing operations section (monthly count, handover session) is expected but may be deferred.

Endpoints called:

```
GET /branch-manager/fixed-assets/overview
```

- Backend must return `FixedAssetsOverviewResponse`. Currently hardcoded in the UI.

5.2 Search Screen

Flutter class: `FixedAssetsSearchScreen`

- Two search modes: General Search and Quick Search.
- General Search requires: Zone, Asset Type, Responsible Employee, and Date.
- Quick Search uses a preset key; only `quick_search_key` is required.
- An 'Upload Image' button exists in the footer but is not yet wired to any backend call.

Endpoints called:

```
POST /branch-manager/fixed-assets/assets/search
```

- If `search_type = general`, all four filter fields are mandatory. If `search_type = quick`, only `quick_search_key` is required.

5.3 Search Results Screen

Flutter class: `FixedAssetsSearchResultScreen`

- Shows summary counts per status (excellent / need_attention / problem).
- Lists assets with multi-select and bulk action support.

- Per-item actions: Settings (location + image), Transfer, Disposal request, Modify Status.
- Preview button exists but currently has no action wired.

Endpoints called:

POST /branch-manager/fixed-assets/assets/search (same as search)

- Response must include nested data.summary and data.items. Asset preview endpoint is needed for the Preview button.

5.4 Receive Assets Screen

Flutter class: [FixedAssetsReceiveAssetsScreen](#)

- Loads a list of assets pending receipt by this branch.
- User selects items and fills in per-item details before confirming.
- Each selected item requires: assigned zone, asset type, total count, per-condition counts, and a receipt photo.
- Only selected items are included in the confirmation request.

Endpoints called:

GET /branch-manager/fixed-assets/receive-assets
POST /branch-manager/fixed-assets/receive-assets/confirm

- The sum of (excellentCount + needAttentionCount + problemCount) should equal assetCount. Backend should validate this business rule.

5.5 Transfers & Disposal Screen

Flutter class: [FixedAssetsTransfersAndDisposalScreen](#)

- Supports three request types: Transfer to Branch, Disposal, and External Transfer.
- Transfer to Branch: requires recipient branch, per-asset transfer reason, and documentation photo.
- Disposal: requires disposal date/time, per-asset disposal reason, condition description, and visual evidence photo.
- External Transfer: only requires type and assets array (approval note shown in UI but not sent).
- Two acknowledgment checkboxes are required by the UI before enabling submit, but acknowledgments are not currently in the request payload.

Endpoints called:

POST /branch-manager/fixed-assets/requests/transfer-disposal

■ Mixed field casing exists in this request (camelCase vs snake_case). Standardization is strongly recommended. See Section 7.

5.6 Requests Screen (Paginated Tabs)

Flutter class: `FixedAssetsRequestsScreen`

- Three paginated tabs: Modifications, Transfers, and Disposals.
- Each tab independently paginates its own list.
- All lists use the standard `PaginationMetaResponse` schema.

Endpoints called:

```
GET /branch-manager/fixed-assets/requests/modifications?page=1
GET /branch-manager/fixed-assets/requests/transfers?page=1
GET /branch-manager/fixed-assets/requests/disposals?page=1
```

■ Page parameter defaults to 1 if omitted.

5.7 Modification Details Screen

Flutter class: `FixedAssetsModificationDetailsScreen`

- Loads full modification request details by ID.
- Displays asset snapshot, modification reason, required actions, and a timeline.
- Timeline shows chronological lifecycle events for the request.

Endpoints called:

```
GET /branch-manager/fixed-assets/requests/modifications/{requestId}
```

■ Must return `ModificationDetailsResponse` including a `timelines` array.

5.8 Modify Asset Status Screen

Flutter class: `FixedAssetsModifyAssetStatusScreen`

- Allows a branch manager to submit a modification request for an asset's status.
- Required inputs: new status, reason, documentation photo, done actions (at least one), next action, approval note.
- KNOWN BUG: `asset_id` is currently sent as an empty string by the UI. This must be fixed before integration.

Endpoints called:

```
POST /branch-manager/fixed-assets/requests/modifications (multipart)
```

■ Backend must reject requests with an empty `asset_id`. Frontend fix is required simultaneously.

6. REST API Endpoints

6.1 Reference Data Endpoints

These lightweight GET endpoints return lookup lists used to populate dropdown filters and pickers across multiple screens.

GET `/branch-manager/fixed-assets/zones`

Returns all zones available in this branch. Used by: Search filters, Receive Assets, Update Settings.

Response Object: ZonesListResponse

GET `/branch-manager/fixed-assets/types`

Returns asset type categories. Used by: Search filters, Receive Assets.

Response Object: TypesListResponse

GET `/branch-manager/fixed-assets/employees`

Returns employees eligible to be assigned as asset custodians. Used by: Search filters (Responsible bottom sheet).

Response Object: EmployeesListResponse

GET `/branch-manager/fixed-assets/branches`

Returns list of branches. Used by: Transfers & Disposal (recipient branch picker).

Response Object: BranchesListResponse

GET `/branch-manager/fixed-assets/assets`

Returns all branch assets. Supports optional `?search=` query param to filter by asset name. Used by: Transfers & Disposal asset picker.

Response Object: AssetsListResponse

Query Parameters:

Parameter	Type	Required	Description
<code>search</code>	string	optional	Filter assets by name substring. Case-insensitive recommended.

6.2 Search Endpoints

POST `/branch-manager/fixed-assets/assets/search`

Full-featured asset search supporting both General and Quick search modes. Returns summary counts alongside the matching asset list.

Request Object: AssetsSearchRequest

Field	Type	Required	Description / Validation Rules
<code>search_type</code>	string	Yes	Mode selector: general or quick.
<code>zone_id</code>	string	required if general	Must be a valid zone id from /zones.
<code>type_id</code>	string	required if general	Must be a valid type id from /types.
<code>employee_id</code>	string	required if general	Must be a valid employee id from /employees.
<code>date</code>	ISO-8601	required if general	Filter date. Date-only or full datetime accepted.
<code>quick_search_key</code>	string	required if quick	One of: update_today, update_yesterday, not_update_1_month, not_update_2_month, in_my_custody, under_maintenance.

Response Object: SearchResponse

■ Validation rule: If `search_type` = general, all four general fields (`zone_id`, `type_id`, `employee_id`, `date`) are required. If `search_type` = quick, only `quick_search_key` is required. Return 422 Unprocessable Entity with field-level errors otherwise.

POST `/branch-manager/fixed-assets/assets/search-image`

Image-based asset search. The UI exposes an 'Upload Image' button that is not yet wired. This endpoint must be implemented to support future image recognition search.

Request: multipart/form-data

Field	Type	Required	Description / Validation Rules
<code>image</code>	file	Yes	Image file only. Max size determined by backend limits.

Response Object: SearchResponse (same shape as text search)

■ This endpoint is currently unimplemented in both frontend and backend.

6.3 Receive Assets Endpoints

GET `/branch-manager/fixed-assets/receive-assets`

Returns a list of assets pending receipt for this branch. Screen loads this list on mount and allows the manager to select items to confirm.

Response Object: ReceiveAssetsListResponse

```
{
  "data": [
    {
      "id": "string",
      "assetName": "string",
      "assetCode": "string",
      "assetImage": "string"
    }
  ]
}
```

POST /branch-manager/fixed-assets/receive-assets/confirm

Confirms receipt of one or more selected assets. Each confirmed item includes zone assignment, asset type, count breakdown by condition, and a receipt photo.

Request: multipart/form-data

Top-level fields:

Field	Type	Required	Description / Validation Rules
type	string	Yes	Receive session type: from_branch or from_finance. Sent as a top-level field, not nested under items.

Per-item fields (one set per selected asset):

Field	Type	Required	Description / Validation Rules
items[assetId][assetId]	string	Yes	Asset ID from ReceiveAssetsItemResponse.id.
items[assetId][assignedZoneId]	string	Yes	Must be a valid zone id from /zones.
items[assetId][assetTypeId]	string	Yes	Must be a valid type id from /types.
items[assetId][assetCount]	int	Yes	Total asset count received. Must be >= 0.
items[assetId][excellentCount]	int	Yes	Count in excellent condition. Must be >= 0.
items[assetId][needAttentionCount]	int	Yes	Count needing attention. Must be >= 0.
items[assetId][problemCount]	int	Yes	Count with problems. Must be >= 0.
items[assetId][image]	file	Yes	Receipt photo. Image file.

■ The type field is a top-level form field and must NOT be nested under items. The frontend uses the assetId value as the array key rather than a numeric index (e.g. items[abc123][assetId]). PHP's \$_FILES parsing handles this but may require custom parsing logic. Backend should validate that excellentCount + needAttentionCount + problemCount equals assetCount.

6.4 Asset Settings (Update Location + Image)

PATCH

/branch-manager/fixed-assets/assets/{assetId}/settings

Updates the zone/location and optionally replaces the image of an existing asset. Triggered from the Settings action on search result items.

Path Parameters:

Parameter	Type	Required	Description
assetId	string	Yes	The unique ID of the asset to update.

Request: multipart/form-data

Field	Type	Required	Description / Validation Rules
new_zone_id	string	Yes	Must be a valid zone id from /zones.
new_image	file	optional	Replacement image. Image files only. Existing image kept if omitted.

6.5 Transfers & Disposal Endpoints

POST `/branch-manager/fixed-assets/requests/transfer-disposal`

Creates a Transfer or Disposal request. The request type drives which conditional fields are required. All three types share a common assets array.

Request: multipart/form-data | Object: TransferOrDisposalRequest

Common Fields (all types):

Field	Type	Required	Description / Validation Rules
type	string	Yes	One of: transfer_to_branch, disposal, external_transfer.

Transfer to Branch — Additional Fields (required when type = transfer_to_branch):

Field	Type	Required	Description / Validation Rules
branchId	string	Yes	Recipient branch ID. Must be a valid branch from /branches.
autoApprove	bool	optional	If true, request is auto-approved without branch review.

Disposal — Additional Fields (required when type = disposal):

Field	Type	Required	Description / Validation Rules
disposalDate	string	Yes	Formatted date string for scheduled disposal.
disposalTime	string	Yes	Localized time string for scheduled disposal.
disposalMethod	string	optional	One of: scrap, sell, return_to_supplier, other.

Assets Array (required for all types):

Field	Type	Required	Description / Validation Rules
assets[i][assetId]	string	Yes	Asset ID. Must be a valid asset in this branch.
assets[i][transferReason]	string	required if transfer_to_branch	Non-empty text.
assets[i][documentationPhotos]	file	required if transfer_to_branch	Single image file.
assets[i][disposalReason]	string	required if disposal	Non-empty text.
assets[i][conditionDescription]	string	required if disposal	Non-empty text.
assets[i][visualEvidence]	file	required if disposal	Single image file.

■ Mixed casing alert: branchId, autoApprove, disposalDate, disposalTime use camelCase while other fields in the system use snake_case. Consider standardising to snake_case and updating the Flutter client accordingly. See Section 7.

POST `/branch-manager/fixed-assets/requests/transfers/{requestId}/approve`

Approves a pending inbound transfer request. No request body required.

Path Parameters:

Parameter	Type	Required	Description
<code>requestId</code>	string	Yes	Valid transfer request ID.

GET `/branch-manager/fixed-assets/requests/transfers/{requestId}`

Returns full details for a specific transfer request including asset list, request metadata, and lifecycle timeline.

Path Parameters:

Parameter	Type	Required	Description
<code>requestId</code>	string	Yes	Valid transfer request ID.

Response Object: `FixedAssetsTransferDetailsResponse` — see Section 4.7 for full shape.

6.6 Modification Request Endpoints

POST `/branch-manager/fixed-assets/requests/modifications`

Submits a new asset status modification request. All fields are mandatory.

Request: `multipart/form-data` | **Object:** `ModifyAssetStatusRequest`

Field	Type	Required	Description / Validation Rules
<code>asset_id</code>	string	Yes	Valid asset ID. KNOWN BUG: UI currently sends empty string. Must be fixed.
<code>new_status</code>	string	Yes	One of: <code>excellent</code> , <code>need_attention</code> , <code>problem</code> .
<code>reason</code>	string	Yes	Non-empty reason text.
<code>attachment</code>	file	Yes	Documentation photo.
<code>done_actions[]</code>	string[]	Yes	Array. Must include at least one of: <code>initial_check</code> , <code>cleaning</code> .
<code>next_action</code>	string	Yes	One of: <code>need_technician</code> , <code>halt</code> , <code>replace</code> .
<code>approval_request_owner_note</code>	string	Yes	Non-empty approval note.

■ **CRITICAL:** `asset_id` is currently sent as an empty string by the Flutter UI. The backend must validate and reject empty `asset_id` with a clear error response. The mobile team must simultaneously fix the UI to pass the selected asset's ID.

GET `/branch-manager/fixed-assets/requests/modifications`

Returns paginated list of modification requests for this branch manager.

Query Parameters:

Parameter	Type	Required	Description
<code>page</code>	int	optional	Page number. Defaults to 1 if omitted.

Response Object: `ModificationRequestsListResponse`

GET

/branch-manager/fixed-assets/requests/modifications/{requestId}

Returns full details for a specific modification request.

Path Parameters:

Parameter	Type	Required	Description
requestId	string	Yes	Valid modification request ID.

Response Object: ModificationDetailsResponse

6.7 Transfer Request Endpoints

GET

/branch-manager/fixed-assets/requests/transfers

Returns paginated list of transfer requests.

Query Parameters: page (int, optional, defaults to 1)

Response Object: TransfersRequestsListResponse

6.8 Disposal Request Endpoints

GET

/branch-manager/fixed-assets/requests/disposals

Returns paginated list of disposal requests.

Query Parameters: page (int, optional, defaults to 1)

Response Object: DisposalsRequestsListResponse

GET

/branch-manager/fixed-assets/requests/disposals/{requestId}

Returns full details for a specific disposal request including asset list, disposal metadata, and lifecycle timeline.

Path Parameters:

Parameter	Type	Required	Description
requestId	string	Yes	Valid disposal request ID.

Response Object: FixedAssetsDisposalsDetailsResponse — see Section 4.8 for full shape.

6.9 Asset Details Endpoint

GET

/branch-manager/fixed-assets/assets/{assetId}

Returns full details for a single asset. Intended to power the Preview action on search result items (currently no action is wired in the UI).

Path Parameters:

Parameter	Type	Required	Description
assetId	string	Yes	Valid asset ID.

Response Object: AssetDetailsResponse

```
{
  "data": {
    "id": "string",
    "name": "string",
    "code": "string",
    "image": "string",
    "location": { "id": "string", "name": "string" },
    "status": "excellent | need_attention | problem",
    "assigned_to": "string",
    "age": "string",
    "value": "string",
    "custody": "string"
  }
}
```

6.10 Dashboard Overview Endpoint

GET `/branch-manager/fixed-assets/overview`

Returns aggregate asset counts for the dashboard summary card. Currently hardcoded in the UI and must be replaced with live data.

Response Object: FixedAssetsOverviewResponse

```
{
  "total": 84,
  "excellent": 78,
  "maintenance": 5,
  "problem": 1
}
```

7. Integration Concerns & Known Issues

The following issues were identified during the API contract analysis. Each concern should be reviewed and resolved jointly by the mobile and backend teams before development begins.

HIGH Empty `asset_id` in Modify Asset Status

The Flutter UI currently sends `asset_id` as an empty string when creating a modification request. The backend must enforce a non-empty, valid asset ID and return a 422 error. The mobile team must simultaneously fix the UI to inject the selected asset's ID into the request.

HIGH Nested Response Shape for Search

The Search endpoint response must follow the exact shape: `{ asset_type_name, data: { summary: {...}, items: [...] } }`. The UI parses `data.summary` and `data.items` as separate blocks. Any deviation from this nesting will break the search results screen.

MEDIUM Mixed Field Casing in Transfers & Disposal

The Transfer/Disposal request uses camelCase for some fields (`branchId`, `autoApprove`, `disposalDate`, `disposalTime`) while the rest of the API uses snake_case. This inconsistency should be standardised. Either update the PHP backend to accept both casings, or coordinate with the mobile team to switch to snake_case.

MEDIUM Receive Assets Array Key Format + top-level type field

The Receive Assets confirmation request now includes a top-level type field (`from_branch` or `from_finance`) that must be sent separately from the items array. Additionally, the frontend uses the `assetId` value as the PHP array key (e.g. `items[abc123][assignedZoneId]`) rather than a numeric index. This is non-standard and may require custom PHP parsing. The backend team should validate that this format is correctly handled.

MEDIUM Acknowledgments Not Sent in Payload

The Transfer/Disposal screen requires two acknowledgment checkboxes to be checked before submission, but these boolean flags are not included in the current request payload. If the backend needs to log or validate acknowledgments, additional fields must be added and the Flutter client must be updated.

LOW Pagination Meta Required on All Lists

All paginated list responses must include the meta block: `{ current_page, last_page, total }`. Missing or incorrectly shaped meta will break the infinite scroll / pagination UI on the Requests tabs.

LOW

Image Fields May Be Empty String

Several response objects include image fields that may have no value. The UI expects an empty string ("") rather than null for missing images. Returning null may cause null-pointer errors in the Flutter client.

8. Suggested Improvements

The following improvements are recommended before or during backend implementation to improve API quality, maintainability, and mobile team developer experience.

1. Standardise all keys to snake_case

Align all request and response field names to snake_case. The Transfer/Disposal request currently contains camelCase fields (branchId, autoApprove, disposalDate, disposalTime) that conflict with the rest of the API. A consistent convention reduces integration bugs.

2. Fix asset_id in Modify Asset Status

Coordinate with the mobile team to ensure the selected asset's ID is correctly passed in the modification request. Backend must enforce validation. This is a blocking issue.

3. Implement Asset Preview/Details Screen

The search result Preview button exists in the UI but has no endpoint wired. Implementing GET /assets/{assetId} and connecting the Preview action will eliminate a dead-end user experience.

4. Add name/code to Search Request

The search screen has a search bar (text input) but no corresponding field in the current API contract. A search_query parameter should be added to AssetsSearchRequest so the user's typed input is actually filtered on the backend.

5. Validate Condition Count Sum in Receive Assets

Add a server-side rule that validates $\text{excellentCount} + \text{needAttentionCount} + \text{problemCount} == \text{assetCount}$. This prevents data integrity issues where totals don't balance.

6. Consider Realtime Updates for Requests Lists

The Requests screen (modifications/transfers/disposals) currently polls on load. For a future iteration, consider WebSocket or Server-Sent Events to push status changes in real time. Not required for v1.

7. Standardise Transfer/Disposal Array Indexing

Consider switching the Receive Assets array key from asset-ID-as-key to numeric indexing (items[0][assetId], items[1][assetId]) for better PHP compatibility.

9. Missing / Unimplemented Endpoints

The following endpoints were identified as missing — either the UI exposes a button or action with no backend wired, or the current implementation is hardcoded. Items marked RESOLVED have been defined in this document revision.

Feature	Proposed Endpoint	Status & Notes
Image-based Search	<code>POST /branch-manager/fixed-assets/assets/search-image</code>	PENDING — Upload Image button exists in Search screen footer but is unimplemented in both frontend and backend.
Asset Preview/Details	<code>GET /branch-manager/fixed-assets/assets/{assetId}</code>	PENDING — Preview action on search result items has no wired endpoint.
Transfer Request Approval	<code>POST /branch-manager/fixed-assets/requests/transfers/{requestId}/approve</code>	PENDING — Approve button visible on pending transfers but endpoint not yet wired.
Transfer Request Details	<code>GET /branch-manager/fixed-assets/requests/transfers/{requestId}</code>	RESOLVED — Response shape fully defined as <code>FixedAssetsTransferDetailsResponse</code> (see Section 4.7).
Disposal Request Details	<code>GET /branch-manager/fixed-assets/requests/disposals/{requestId}</code>	RESOLVED — Response shape fully defined as <code>FixedAssetsDisposalsDetailsResponse</code> (see Section 4.8).
Dashboard Overview Stats	<code>GET /branch-manager/fixed-assets/overview</code>	PENDING — Dashboard counts are hardcoded in the UI. Must be replaced with live API data.

10. Endpoint Quick Reference

Complete index of all proposed Fixed Assets API endpoints. All URLs are relative to the base path `/branch-manager/fixed-assets`.

Method	Endpoint (relative)	Auth	Body	Description
GET	/overview	Bearer	—	Dashboard asset summary counts
GET	/zones	Bearer	—	List all zones
GET	/types	Bearer	—	List all asset types
GET	/employees	Bearer	—	List responsible employees
GET	/branches	Bearer	—	List all branches
GET	/assets	Bearer	?search=	List/search assets (picker)
GET	/assets/{assetId}	Bearer	—	Single asset details
PATCH	/assets/{assetId}/settings	Bearer	multipart	Update asset zone + image
POST	/assets/search	Bearer	JSON	Search assets (general or quick)
POST	/assets/search-image	Bearer	multipart	Search by uploaded image
GET	/receive-assets	Bearer	—	List pending receive assets
POST	/receive-assets/confirm	Bearer	multipart	Confirm asset receipt
POST	/requests/transfer-disposal	Bearer	multipart	Create transfer or disposal request
GET	/requests/modifications	Bearer	?page=	List modification requests (paginated)
POST	/requests/modifications	Bearer	multipart	Create modification request
GET	/requests/modifications/{requestId}	Bearer	—	Modification request details
GET	/requests/transfers	Bearer	?page=	List transfer requests (paginated)
GET	/requests/transfers/{requestId}	Bearer	—	Transfer request details
POST	/requests/transfers/{requestId}/approve	Bearer	—	Approve a transfer request
GET	/requests/disposals	Bearer	?page=	List disposal requests (paginated)
GET	/requests/disposals/{requestId}	Bearer	—	Disposal request details